CONSTRUCTION SITE SAFETY-ANALYZER AND ENHANCER USING OPTIMIZED YOLOV8 AND R-CNN

PROJECT REPORT

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Construction sites pose significant safety risks, often due to non-compliance with personal protective equipment (PPE) requirements. Traditional manual supervision methods are inefficient and prone to errors. This research presents the Construction Site Safety Analyzer, an AI-driven system that automates PPE detection using YOLOv8 and Region-Based CNN (R-CNN). The system analyses live video feeds to identify workers and their protective gear, ensuring real-time compliance monitoring. When a violation is detected, a GSM module sends immediate alerts to supervisors for prompt action. By leveraging deep learning and IoT, the system enhances workplace safety while reducing human oversight errors.

Designed for adaptability, it can be deployed on edge devices or cloud-based platforms, making it suitable for diverse construction environments. Performance is evaluated using precision, recall, and F1-score metrics, demonstrating high accuracy in detecting PPE violations. Experimental results confirm that the system effectively minimizes manual supervision efforts, strengthens safety enforcement, and mitigates accident risks. Future improvements may include enhanced detection in low-light conditions and faster processing through edge computing. This approach provides a smart, automated solution for improving construction site safety.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
CNN	Convolutional Neural Network
IOT	Internet Of Things
IDE	Integrated Development Environment
PPE	Personal Protective Equipment
SMS	Short Message Service
GPU	Graphical Processing Unit
YOLO	You Only Look Once
GSM	Global System for Mobile Communications
CCTV	Closed Circuit Television
R-CNN	Region-Based Convolutional Neural Network
IOU	Intersection Over Union

CHAPTER I
INTRODUCTION

1.1 INTRODUCTION

Construction sites are high-risk areas where ensuring worker safety is crucial. Employees are required to wear personal safety wears such as helmets, vests, and masks to lower the risk of injuries. However, present enforcement of PPE monitoring method is time-consuming, inconsistent, and prone to human error. Traditional monitoring methods rely on physical inspections, which may not be effective in large and dynamic construction areas.

This research introduces a Construction Site Safety Analyzer that leverages YOLOv8 and Region-Based CNN (R-CNN) to detect safety compliance in real time. The system processes live video feeds to identify workers and verify the presence of required safety gear. In case of a violation, a GSM module is used to send immediate alerts to site supervisors, allowing for timely intervention.

Construction sites are often divided into different zones based on activity type and risk level, such as general work areas, restricted zones, and high-risk equipment zones. Effective monitoring of these areas is essential for maintaining safety standards, ensuring adherence to regulations, and preventing workplace hazards. By integrating deep learning with IoT-based alert systems, this research aims to enhance workplace safety, reduce reliance on manual inspections, and create a more efficient system for PPE compliance enforcement.

Recent developments in artificial intelligence (AI) have opened up new possibilities for improving safety in building sites. Construction sites can streamline surveillance of worker presence, equipment usage, PPE legal compliance and other safety-critical aspects inside the work area by implementing AI-powered object identification models[20][19]. By enabling early risk identification and prompt alerts to stop accidents, such technology provides a proactive approach to safety that eventually promotes a safer and more effective workplace.

1.2 SCOPE OF THE WORK

Traditional PPE monitoring in construction sites relies on manual inspections, which are often ineffective and error-prone. This study introduces an AI-powered Construction Site Safety Analyzer using YOLOv8 and R-CNN for real-time PPE detection. The system processes CCTV feeds to identify workers and verify compliance with safety gear. In case of a violation, a GSM module sends instant alerts to supervisors for quick action.

By automating PPE detection, the system enhances safety compliance, reduces human errors, and improves site security. Designed for scalability, it can be deployed on edge devices or cloud platforms for real-time processing. Future improvements may include multi-sensor integration and enhanced detection in low-light conditions.

CHAPTER II
LITERATURE REVIEW / EXISTING SYSTEM

2.1 ANALYSIS OF EXISITING LITERATURE

- **Title**: Construction Site Safety Management: A Computer Vision and Deep Learning Approach

  Authors: Jaekyu Lee, Sangyub Lee

  Year of Publication: 2023

  Description: This image recognition technology to study various alternatives to enhancing the safety of construction workers. Three object recognition scenarios for safety on a construction site were created, and a respective object recognition model was formulated for each scenario. The first object recognition model verifies whether there are any construction workers at the site. The second object recognition model checks the risk of falling (falling from a structure or falling) when working at a height. The third object recognition model checks if the workers are properly wearing safety helmets and vests. These three models were newly developed using the image data gathered from the construction sites and synthetic image data gathered from the virtual environment based on transfer learning. Specifically, we validated an artificial intelligence model from a virtual environment in this research. Therefore, simulating and conducting tests on worker falls and fall injuries, which are hard to re enact by humans, are effective algorithm verification techniques.

  Title: Multi-Task Intelligent Monitoring of Construction Safety Based on Computer Vision

  Authors: Lingfeng Liu, Zhigang Guo, Zhengxiong Liu, Yaolin Zhang, Ruying Cai, Xin Hu, Ran Yang and Gang Wang

Year of Publication: 2024

Description: This paper investigates the application of deep learning models, including computer vision and Computer Vision (CV), for detecting various types of PPE in construction. By training on a dataset containing hard hats, masks, and safety vests, the model improves detection in complex settings. Faster CNN's high accuracy in detecting partially obscured objects is noted, though the authors recognize the trade-off with processing speed.

2.2 CHALLENGES IDENTIFIED

The "Personal Protective Equipment Detection for Construction" paper highlights several challenges that reduce the effectiveness of automated site safety analyzer systems. One of the primary challenges is the variation in worker appearance and PPE types, as different brands, colors, and designs of helmets, vests, and masks can lead to inconsistent detection results. Apart from this, environmental factors such as fluctuating lighting conditions, shadows, and extreme weather can negatively affect the accuracy of the system, making it difficult to distinguish PPE from surrounding objects. The positioning of cameras also plays a crucial role, as improper angles or occlusions caused by other workers, equipment, or structural elements can result in missed detections. Another major concern is real-time performance, as high-resolution video streams demand substantial computational power, leading to potential latency issues in processing and alert generation.

Furthermore, false positives and false negatives present significant challenges, as objects with similar colors or shapes to PPE may be mistakenly classified, while partially visible or improperly worn PPE might go undetected. The system's integration into an operational construction site is another hurdle, as hardware limitations, unstable network connectivity, and

the need for continuous monitoring increase deployment complexity. Additionally, worker compliance and behavior can impact detection efficiency, as some employees may intentionally remove or misuse safety gear, making it harder for the system to enforce compliance effectively.

2.3 OBJECTIVES

The objective of a safety monitoring system using Convolutional Neural Networks (CNN) and computer vision is to create an automated, real-time solution that accurately detects essential safety gear such as helmets, vests, and gloves, ensuring compliance with safety standards in industrial environments. By leveraging CNN's strong feature extraction capabilities, the system can analyse video feeds or images from surveillance cameras to identify safety equipment with high precision, even in complex or dynamic settings. The integration of computer vision enables the system to operate continuously and autonomously, minimizing the need for manual inspections and reducing human error.

The system seeks to achieve high accuracy in detecting safety gear across diverse environments, handling challenges such as varying lighting conditions, occlusions, and different viewing angles. By using CNN models optimized for object detection, it provides reliable performance, enabling comprehensive monitoring of compliance without disrupting workflow. Furthermore, the automated nature of the system ensures continuous, non-intrusive monitoring, making it suitable for large-scale deployment in high-risk areas where safety is paramount.

Additionally, the system aims to provide immediate alerts for safety violations, allowing for prompt intervention and reducing the probability of accidents. This approach not only enhances worker protection but also

streamlines the safety monitoring process, making it scalable and adaptable for various industrial applications, from construction sites to manufacturing plants.

CHAPTER III
REQUIREMENT SPECIFICATION

3.1 REQUIREMENT SPECIFICATIONS

The Construction Site Safety Analyzer is designed to ensure PPE compliance using deep learning and IoT-based alert systems. The system requires real-time video processing, high-accuracy object detection, and an automated notification mechanism to enhance workplace safety. It integrates YOLOv8 and Region-Based CNN (R-CNN) to detect helmets, vests, and masks from CCTV footage. The software is developed using Visual Studio, ensuring smooth integration with hardware components such as CCTV cameras and GSM modules for real-time alerts. A GPU-enabled system is required for efficient deep learning inference. The solution aims to minimize manual supervision errors, improve safety enforcement, and enable automated monitoring in construction environment[1][14].

3.2 MODULES DESCRIPTION

Input Module: This module captures visual data, such as images or videos from cameras installed at the construction site. The video feed is continuously streamed to the system for analysis [1][17].

Key Components:

- Camera devices (CCTV or mobile cameras)
- Data pre-processing (frame extraction for video input)
- Input pipelines for image or video streaming

Preprocessing Module: This module prepares the captured images or video frames for further processing. It applies techniques like resizing, normalization, and noise reduction to optimize the images for computer vision[1][15].

Key Components:

- Image normalization (scaling, color correction)
- Noise reduction (filters)
- Technologies Used: NumPy

PPE Detection Module: Uses YOLOv8 and R-CNN to identify helmets, vests, and masks in real-time. It processes each video frame to detect compliance with safety regulations[1][22][23].

Key Components:

- Deep Learning models like R-CNN
- YOLOv8

Violation Detection And Alert Module: Analyzes detection results to check for PPE violations. If a worker is non-compliant, it sends instant alerts to supervisors via the GSM module.

Key Components:

- GSM Modules

3.3 ALGORITHMS USED

3.3.1 Region Based Convolutional Neural Network Algorithm:

R-CNN follows a two-step approach, first identifying potential object regions (region proposals) and then classifying them which is depicted in the below given **Figure 3.1**. It extracts detailed object features, making it useful for refining detections and reducing false positives. In this system, R-CNN works alongside YOLOv8 to improve detection accuracy, especially in complex conditions like poor lighting or occlusion[1][10].

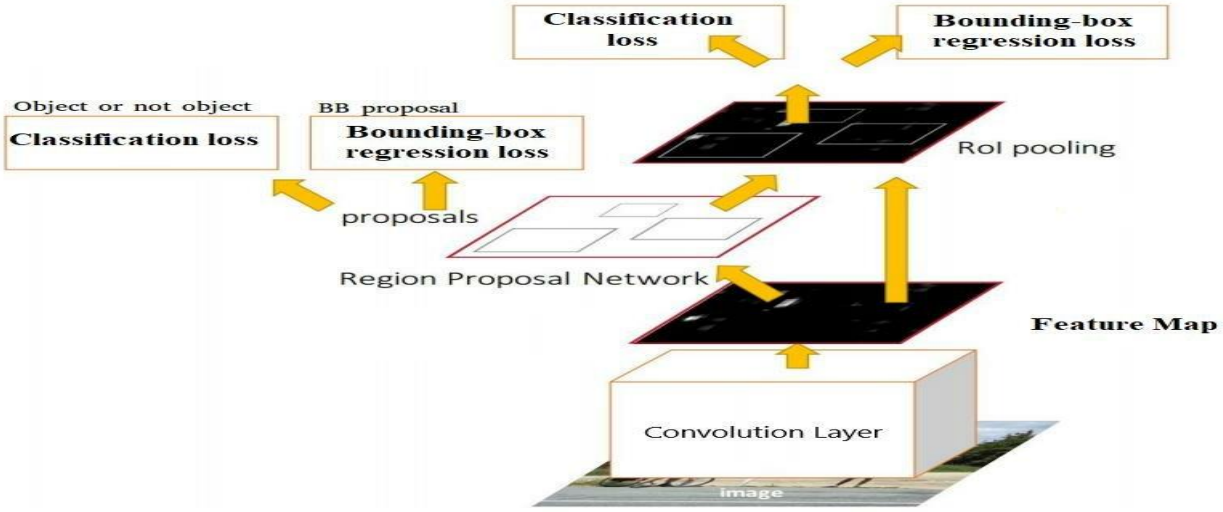


Figure 3.1 Workflow of R-CNN

3.3.2 YOU ONLY LOOK ONCE VERSION-8 (YOLOv8):

YOLOv8 is a deep learning-based object detection model optimized for speed and accuracy. It divides an image into a multiple small grid and predicts bounding boxes along with confidence scores in a single pass, making it highly efficient. The model is trained to recognize PPE such as helmets, vests, and masks, ensuring real-time detection in construction environments[23][24]. The detailed illustration is shown in below **figure 3.2.**

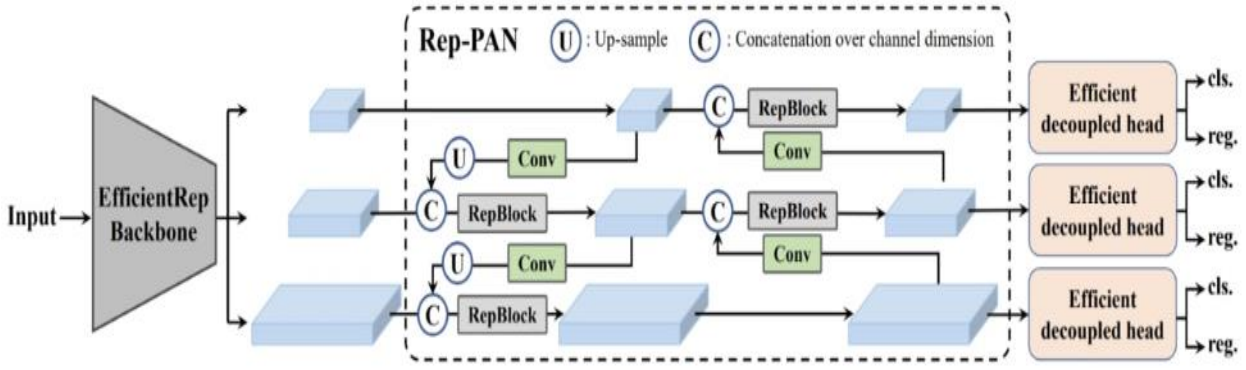


Figure 3.2 Architecture of YOLOv8: Feature Extraction and Detection Pipeline

CHAPTER IV
EXPERIMENTAL WORK

4.1 IMPLEMENTATION METHODS

4.1.1 Data Collection and Annotation

High-quality datasets containing images of construction workers with and without PPE (helmets, vests, and masks) are gathered from various sources and labeled. Then the Labeled data is then used for training the object detection model[1][2].

4.1.2 Data Preprocessing

Prepare the dataset to ensure efficient and accurate processing by Faster R-CNN. Techniques like rotation, scaling, brightness adjustments, and flipping are applied to increase model robustness. These preprocessing steps help improve the performance under different conditions commonly encountered on construction sites. An augmented, standardized dataset ready for efficient model training and increased resilience to diverse construction site conditions[3].

4.1.3. Object Detection using YOLOv8

The YOLOv8 model is employed for real-time detection of PPE compliance. The input is processed through the Efficient Rep Backbone, which extracts required features. These features are further refined using Rep-PAN (Path Aggregation Network) to enhance detection accuracy across different scales. The final Efficient Decoupled Head performs classification (determining the presence of PPE) and regression (localizing bounding boxes)[4][15].

4.1.4. PPE Violation Detection

Once the PPE components are identified, the system analyzes worker compliance. If a worker is detected without essential safety gear, the model flags it as a violation. The confidence score for each detection is calculated based on IoU and threshold values are used to minimize false positives[10].

4.1.5. Real-Time Notification via GSM Module

In case of any safety violations, the system immediately triggers an alert using a **GSM module**. A message is sent to the supervisor, specifying the location and type of violation. This real-time communication ensures quick intervention and improved site safety[1][13].

4.1.6. Performance Evaluation

The model's accuracy is assessed using metrics such as Precision, Recall, F1-Score, and mAP (Mean Average Precision). These metrics help determine the efficiency of PPE detection and the reliability of the notification system. The model is further fine-tuned based on evaluation results to improve overall performance[1][24].

4.1.7. System Integration and Real-Time Deployment Testing

Deploy the model on actual or simulated construction site footage to validate its real-time capabilities. Integrate the model into a real-time monitoring system connected to site cameras, observing its ability to detect violations and send alerts within seconds. Test the alerting system and evaluate detection accuracy when workers are moving, in crowded areas, or when the visibility of gear is partially obscured.

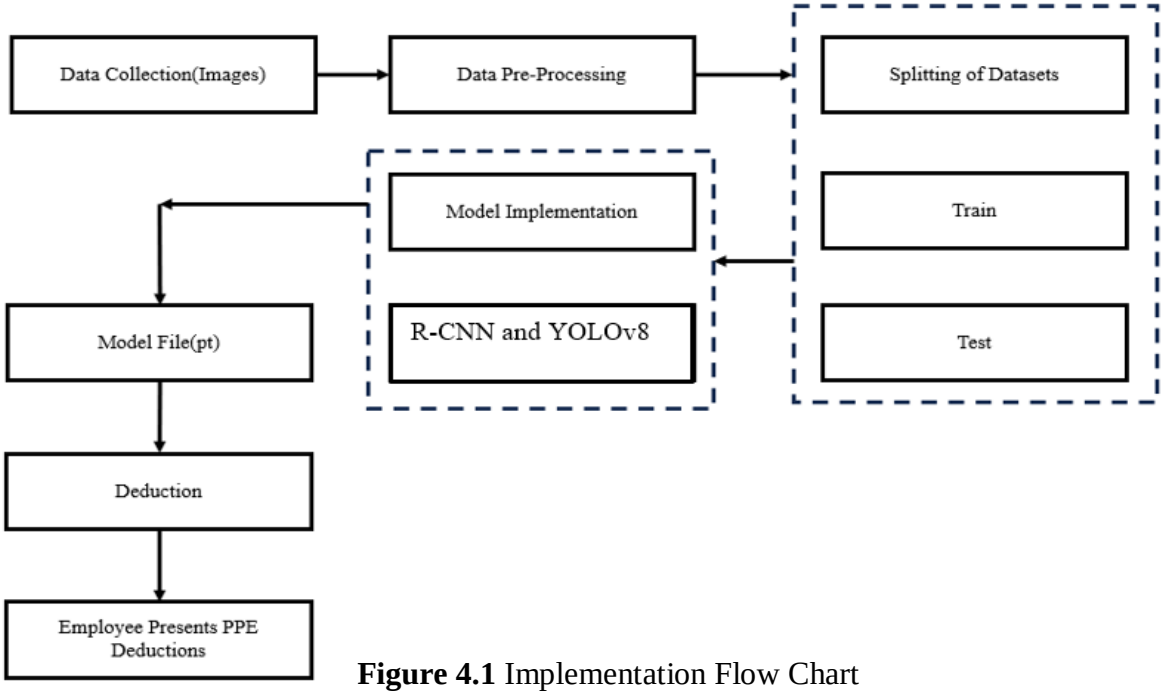


Figure 4.1 Implementation Flow Chart

The above-mentioned **Figure 4.1** illustrates the overall flow of implementation methodology. It begins with collecting and preprocessing image data, followed by dataset splitting for training and testing. A model is implemented using OpenCV and ResNet, producing a model file. The model detects whether employees are wearing PPE, leading to deduction actions if required[1][6].

4.2 ARCHITECTURE DESIGN

The Construction Site Safety Analyzer consists of multiple stages, including feature extraction, object detection, violation assessment, and real-time alerting to ensure accurate monitoring of PPE compliance. The input to the system is a live video stream from CCTV cameras deployed at the construction site. These frames are processed using YOLOv8 object detection model. The Efficient Rep Backbone extracts key features from the images, while the Rep-PAN (Path Aggregation Network) enhances feature

representation for detecting PPE components such as helmets, vests, and masks.

After detection, the system evaluates whether workers are following safety regulations. If non-compliance is detected, the GSM module instantly sends an alert message to the supervisor, allowing for immediate corrective action. By combining deep learning, IoT-based communication, and real-time processing, this architecture enhances workplace safety by reducing manual supervision errors and enabling swift responses to safety violations[1][4]. The Below given **Figure 4.2** depicts the overall architecture of proposed system.

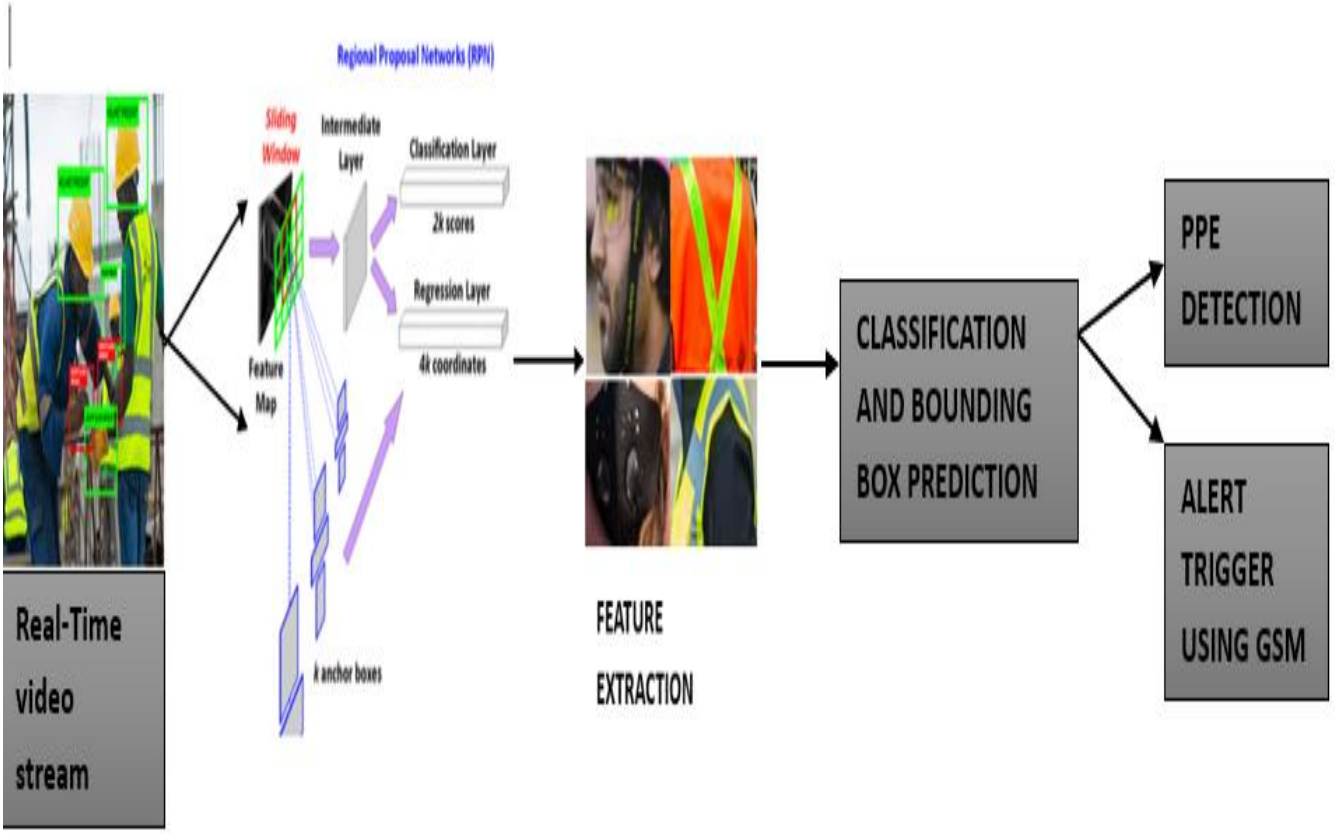


Figure 4.2 Overall Architecture of the System

4.3 SYSTEM DESIGN

4.3.1 INPUT DESIGN

The input design of the Construction Site Safety Analyzer ensures efficient data acquisition and processing for real-time PPE detection. The system receives input from multiple sources, primarily CCTV cameras, which continuously capture live video footage from the construction site. These frames undergo preprocessing before being fed into the YOLOv8 model for object detection. The input design focuses on handling variations in lighting, angles, occlusions, and worker movement, ensuring high detection accuracy[21].

4.3.2 OUTPUT DESIGN

The output design of the Construction Site Safety Analyzer focuses on delivering accurate and real-time results for PPE compliance monitoring. The system processes video feeds and generates outputs in multiple formats to ensure effective safety enforcement. The primary output is a detection report, where workers are identified along with their PPE status whether they are wearing a helmet, vest, and mask or violating safety regulations[20].

When a violation is detected, the system triggers an instant alert via GSM modules, notifying supervisors through SMS or email notifications. Additionally, visual indicators such as bounding boxes are drawn on detected workers in the video feed to highlight compliance status. A log of detected violations is stored in a database, allowing site managers to review compliance trends over time. To enhance usability, the system also supports an interactive dashboard where safety officers can monitor real-time PPE detection results, access historical data, and generate compliance reports. Future enhancements may include automated message alerts.

4.4 UNIFIED MODELLING LANGUAGE

4.4.1 USE CASE DIAGRAM

The below **Figure 4.3** use case diagram illustrates workflow of the Construction Site Safety Analyzer system. Site workers are monitored through live video stream data collection, which includes PPE compliance verification using YOLOv8 and R-CNN algorithms for accurate detection. The system identifies safety gear compliance and detects any violence or unusual activity. In case of violations, it sends an alert call and message via the GSM module to the site supervisors. This ensures real-time monitoring and swift action for improved safety enforcement[15].

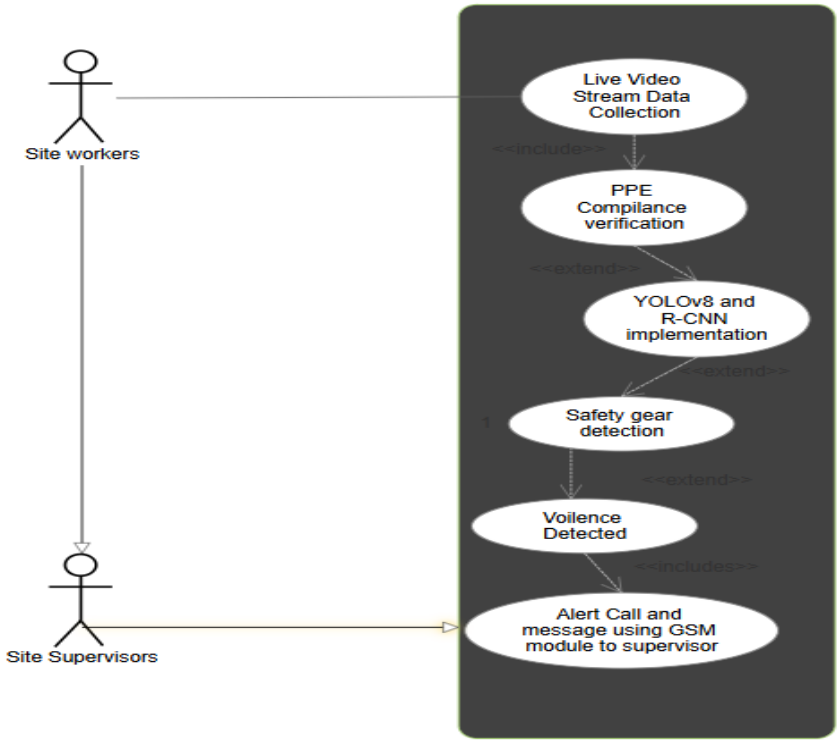


Figure 4.3 Use case Diagram

4.4.2 SEQUENCE DIAGRAM

The below **Figure 4.4** sequence diagram depicts the workflow of the Construction Site Safety Analyzer system. It begins with the user providing real-time video streams , which undergoes preprocessing for noise reduction and enhancement. The image is then analysed by the R-CNN and YOLOv8 model to detect PPE compliance or potential violence. The system performs feature extraction to determine if any safety violations or violent behaviour are present. If detected, an alert is sent via the GSM module to notify the supervisor[16][17].

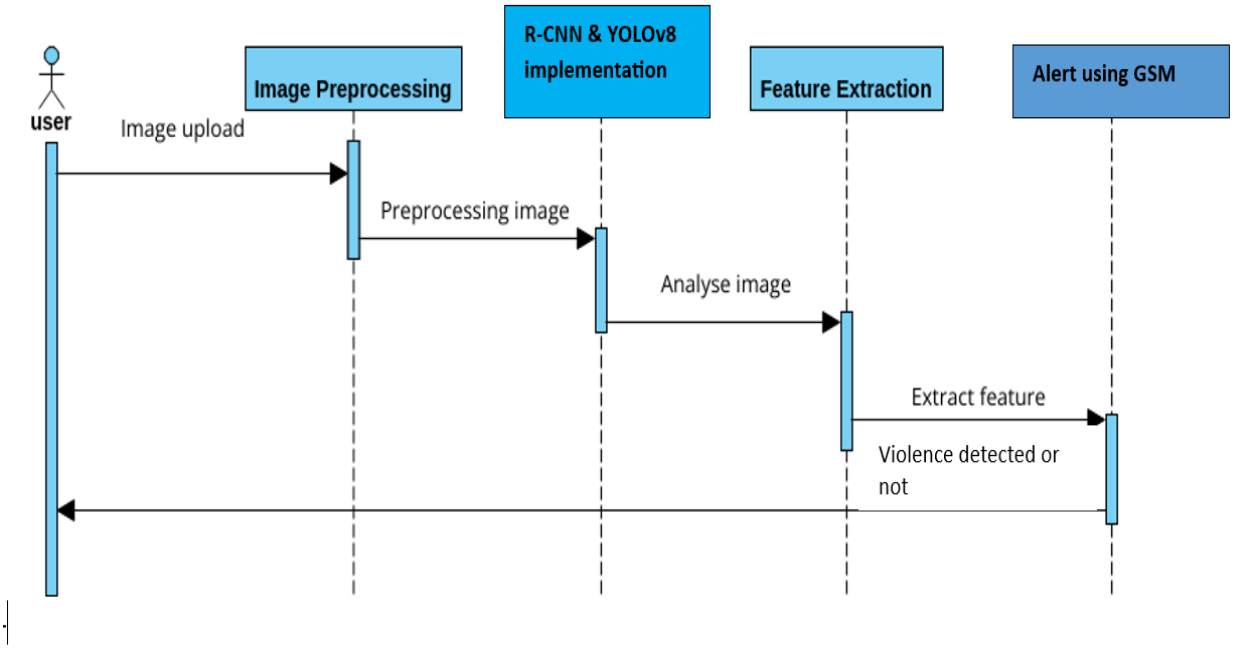


Figure 4.4 Sequence Diagram

CHAPTER V
RESULTS
AND
DISCUSSION

5.1 RESULT ANALYSIS

The Construction Site Safety Analyzer was evaluated based on its ability to accurately detect helmets, vests, and masks in real-time using YOLOv8 and R-CNN models. The system was tested on diverse construction site environments with varying lighting conditions, worker movements, and occlusions to measure its reliability. The results showed a high detection accuracy, with the YOLOv8 model achieving faster processing times compared to R-CNN, making it suitable for real-time applications.

The system effectively identified PPE compliance and flagged violations with minimal false positives. Performance metrics were analysed, showing that the model performed consistently well even in challenging conditions. Additionally, the integration of GSM-based alert notifications ensured immediate response to safety violations, enhancing workplace monitoring efficiency.

The proposed system demonstrated improved detection speed and accuracy over traditional manual supervision methods, reducing human error and ensuring a proactive approach to safety enforcement. Future improvements could focus on multi-camera integration, enhanced night vision capabilities, and AI-driven predictive safety insights to further strengthen PPE compliance monitoring. The below **Table 5.1** depicts the Performance evaluation of proposed system.

Table 5.1 Performance Evaluation Table

PPE TYPE	PRECISION (%)	RECALL (%)	F1-SCORE (%)	DETECTION SPEED (Ms/frame)
Helmet	92.5	89.8	91.1	35
Vest	91.2	88.5	89.8	37
Mask	90.3	87.9	89.1	40
Overall	91.3	88.7	90.0	37.3

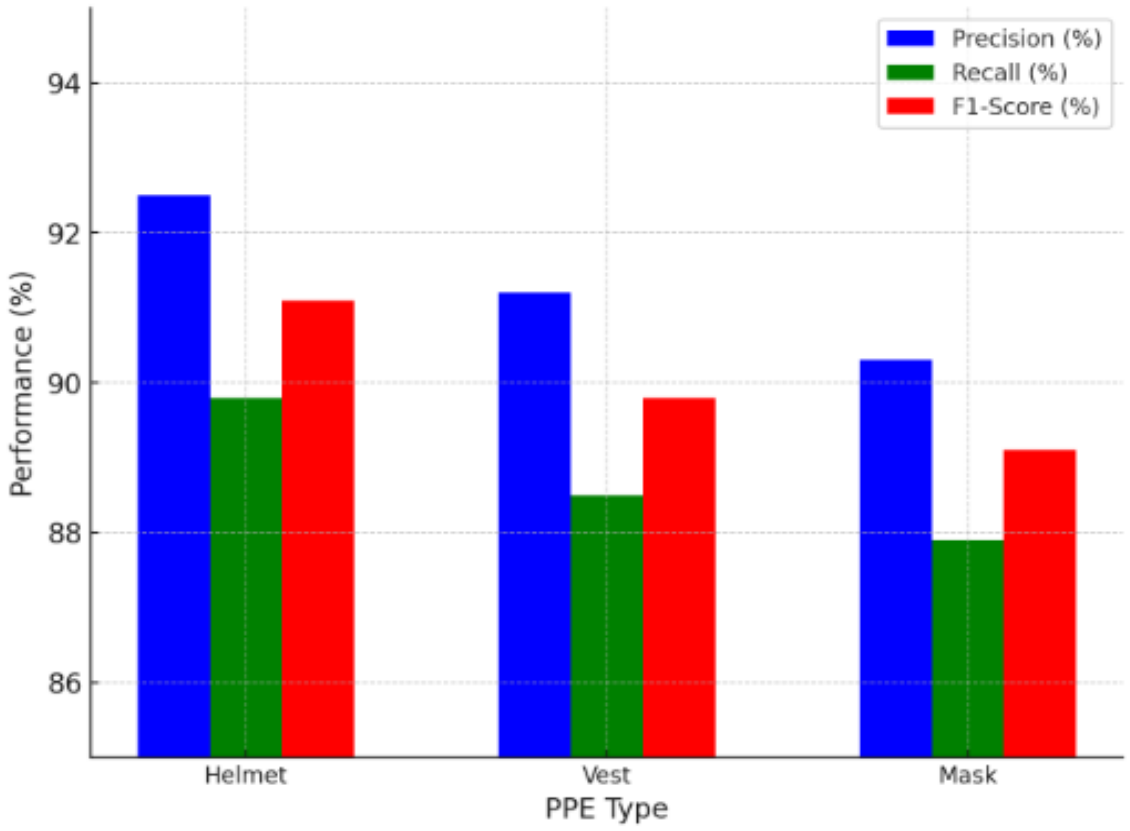


Figure 5.1 Performance evaluation graph

The above **Figure 5.1** illustrates a graph that tracks four key evaluation metrics Precision, Recall, and F1 Score over training epochs whose values are mentioned in above **Table 5.1**. Each metric is represented by a distinct coloured line:

Precision (Blue): Measures the proportion of correctly identified positives among all identified positives. It improves consistently, reaching around 80% by the end, which suggests the model becomes more selective and avoids false positives.

Recall (Green): Represents the ability of the model to correctly identify all true positives. Its gradual increase to approximately75% by the final epochs indicates that the model is becoming better at recognizing relevant cases.

F1 Score (Red): This metric, which is the mean value of Precision and Recall, also improves over time. It reflects the model's balanced performance between precision and recall, showing around 78% by epoch 20.

CHAPTER VI
CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The Construction Site Safety Analyzer effectively enhances PPE compliance monitoring using YOLOv8, R-CNN, and GSM-based alert systems. Real-time detection and instant notifications improve safety enforcement, ensuring that non-compliance is addressed promptly.

The results demonstrate that automated safety monitoring is more efficient, accurate, and scalable than traditional methods. The system successfully operates in diverse environmental conditions, maintaining high detection accuracy and low false alarm rates. Future enhancements could include multi-camera integration, IoT-based tracking, and predictive safety analytics to further optimize workplace safety.

This approach represents a significant step toward smart construction site monitoring, ensuring a safer work environment and reducing the risk of accidents through AI-driven automation.

6.2 FUTURE ENHANCEMENTS

In the future to further improve PPE compliance monitoring, we plan to integrate RFID (Radio Frequency Identification) tags to identify individual workers and link their PPE compliance status to their unique ID. Each worker will be assigned an RFID tag, which the system will detect along with their helmet, vest, and mask status. If a violation occurs, the system will not only flag the non-compliance but also identify the specific worker responsible. This information will be instantly communicated to the supervisor, ensuring targeted corrective action. Additionally, we aim to implement a voice alert system, where the supervisor receives an automated audio notification mentioning the worker’s name and PPE violation. This enhancement will

further streamline safety enforcement by making violation alerts more personalized, faster, and more effective in large construction environments.

ANNEXURE

APPENDIX 1

Source code:main.py

```
import cv2
import torch
import numpy as np
import time
import smtplib
from email.mime.multipart import MIMEMultipart
from email.mime.text import MIMEText
from email.mime.base import MIMEBase
from email import encoders
import serial
arduino=serial.Serial(port='COM6', baudrate=115200, timeout=1)
fromaddr="jabinjoseph321@gmail.com"
toaddr= "balambalganesan13@gmail.com"
def mail(text):
    print(text)
    msg=MIMEMultipart()
    msg['From']=fromaddr
    msg['To']=toaddr
    msg['Subject']="NOT_SAFETY"
    body=text
```

```
    msg.attach(MIMEText(body,'plain'))
    filename="output/img.jpg"
    attachment=open("output/img.jpg","rb")
    p=MIMEBase('application','octet-stream')
    p.set_payload((attachment).read())
    encoders.encode_base64(p)
    p.add_header('Content-Disposition',"attachment;
filename=%s"%filename)
    msg.attach(p)
    s=smtplib.SMTP('smtp.gmail.com',587)
    s.starttls()
    s.login(fromaddr,"rvgbrugvgclgqiql")
    text=msg.as_string()
    s.sendmail(fromaddr,toaddr,text)
    s.quit()
def detect_objects_live(weights_path='best.pt', conf_threshold=0.2):
    # Load YOLOv5 model
    model = torch.hub.load('ultralytics/yolov5', 'custom', path=weights_path,
force_reload=True)
    # Set device
    device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
    model.to(device)
    # Open video capture device (webcam)
```

```
cap = cv2.VideoCapture(0)
while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    # Perform inference
    results = model(frame)
    # Get bounding boxes, confidence scores, and class labels
    boxes = results.xyxy[0]  # Bounding boxes in (x1, y1, x2, y2) format
    confidences = boxes[:, 4]  # Confidence scores
    class_labels = boxes[:, 5]  # Class labels
    # Filter detections based on confidence threshold
    detections_above_threshold = boxes[confidences > conf_threshold]
    # Draw bounding boxes for detections above threshold
    for detection in detections_above_threshold:
        label = int(detection[5])
        score = float(detection[4])
        bbox = detection[:4].cpu().numpy().astype(int)
        cv2.rectangle(frame, (bbox[0], bbox[1]), (bbox[2], bbox[3]), (0, 255,2)
        cv2.putText(frame, f'{model.names[label]}: {score:.2f}',
        (bbox[0], bbox[1] - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)
```

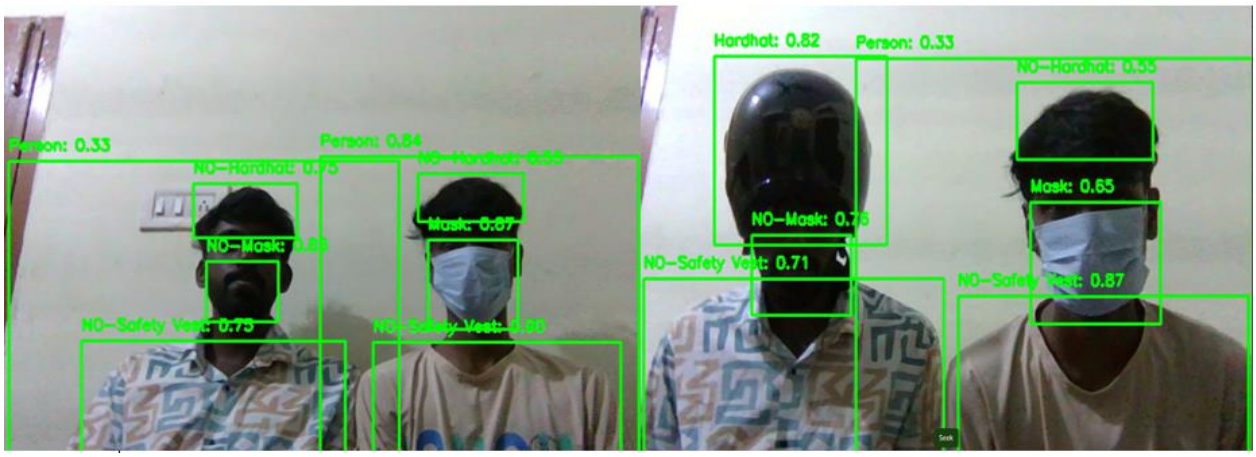
```
    # print(model.names[label])
    klass=model.names[label]
    # NO-Safety Vest,NO-Mask,NO-Hardhat
    if klass == "NO-Hardhat":
        cv2.imwrite('output/img.jpg', frame)
        print("Without Hardhat")
        #mail("NOT_SAFETY")
        arduino.write(b"A")
        print("A Msg send")
    elif klass == "NO-Safety Vest":
        cv2.imwrite('output/img.jpg', frame)
        print("Without Safety Vest")
        #mail("NOT_SAFETY")
        arduino.write(b"B")
        print("B Msg send")
    elif klass == "NO-Mask":
        cv2.imwrite('output/img.jpg', frame)
        print("Without Mask")
        #mail("NOT_SAFETY")
        arduino.write(b"C")
        print("C Msg send")
# Display the frame
cv2.imshow('Safety Detection', frame)
```

```
        # Break the loop on 'q' key press
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break
    # Release resources
    cap.release()
    cv2.destroyAllWindows()
# Example usage
detect_objects_live()
```

APPENDIX 2

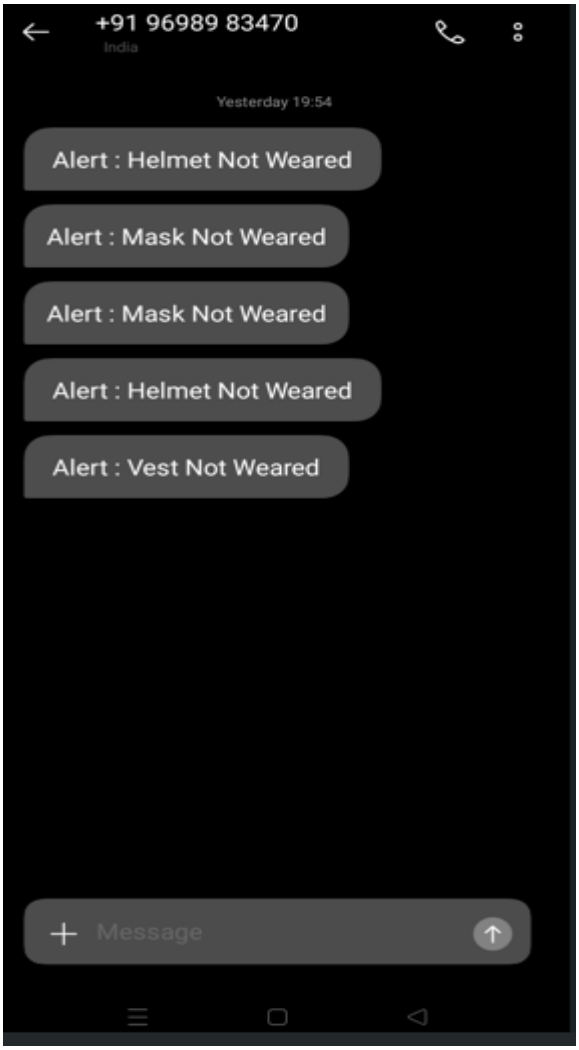
OUTPUT

SCREEN SHOT 1



The above-mentioned screen shot-1 illustrates how the model is being executed in visual studios terminal.

SCREEN SHOT 2



The above given screenshot-2 illustrates the notification or the alert mechanism used in this system. It informs the supervisors whether the environment is safe or not.

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REFERENCES

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